IMPERIAL COLLEGE OF SCIENCE, TECHNOLOGY AND MEDICINE

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EXAMINATIONS 2021

WAVELETS AND APPLICATIONS SOLUTIONS

SOLUTIONS

1. (a) Since the Haar basis is orthogonal, the wavelet coefficients are:

$$\alpha_{m,n} = \langle f(t), \psi_{m,n}(t) \rangle = \int_0^1 \psi_{m,n}(t)dt = \begin{cases} 2^{-m/2} & m \geq 1 \;\; n = 0 \\ \\ 0 & \text{otherwise} \end{cases}$$

(b) Clearly $\|f(t)\| = 1$, moreover,

$$\sum_n \sum_m |\alpha_{m,n}|^2 = \sum_{m=1}^{\infty} 2^{-m} = \frac{1}{1 - 1/2} - 1 = 1$$

So Parseval equality is satisfied.

(c) i. Taking the Fourier transform of the two-scale equation, we get

$$\hat{\varphi}(\omega) = \frac{1}{\sqrt{2}} G_0(e^{j\omega/2}) \hat{\varphi}(\omega/2).$$

ii. Evaluating the two-scale equation at $\omega = 0$ yields $\hat{\varphi}(0) = \frac{1}{\sqrt{2}} G_0(1)\hat{\varphi}(0)$ which implies $G_0(1) = \sum_n g_0[n] = \sqrt{2}$.

iii. Orthogonality implies

$$|G_0(e^{j\omega})|^2 + |G_0(e^{j\omega+\pi})|^2 = 2.$$

Evaluating at $\omega = 0$ yields

$$|G_0(1)|^2 + |G_0(-1)|^2 = 2.$$

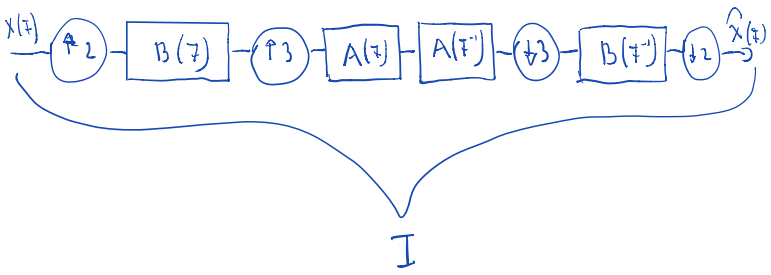
Since $|G_0(1)|^2 = 2$, we have $G_0(-1) = 0$. This implies

$$G_0(-1) = \sum_n (-1)^n g_0[n] = \sum_n g_0[2n] - \sum_n g_0[2n+1] = 0.$$

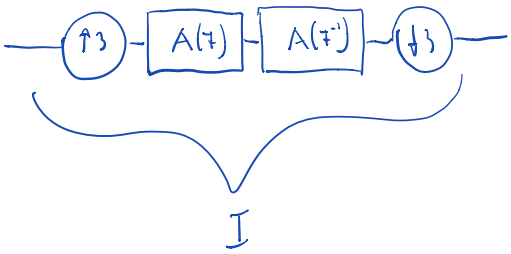
Therefore $\sum_n g_0[2n] = \sum_n g_0[2n+1]$.

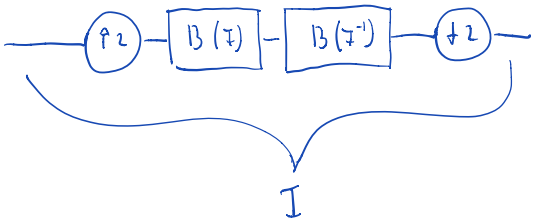
2. (a) This requires application of the Noble Identities. We obtain $M = N = 6$, $C(z) = A(z)B(z^3)$, and $D(z) = C(z^{-1})$.

   (b) The whole system is idempotent if and only if $\langle a[n], a[n-3k] \rangle = \delta_k$. This can be shown by noticing that the system is idempotent if and only if the subsystem shown below is the identity.



This is satisfied when the two following subsystems are themselves identity.





[4/10]

The requirement that the first of these two systems be the identity implies:

$$\frac{X(z)}{3}\left[\sum_{n=0}^{2} A(W_3^n z^{1/3})\right] = X(z)$$

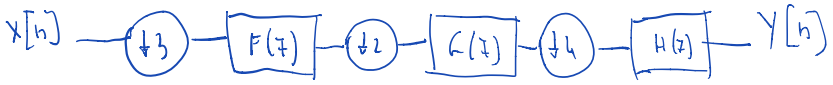
which yields

$$\langle a[n], a[n-3k] \rangle = \delta_k \Leftrightarrow \frac{1}{3}\sum_{n=0}^{2} A(W_3^n z^{1/3}) = 1 \qquad (1)$$

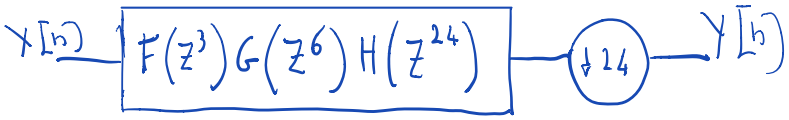
with $W_N^k = e^{-j2\pi k/N}$. The second sub-system is by construction idempotent, since

$$\langle b[n], b[n-2k] \rangle = \delta_k \Leftrightarrow B(z)B(z^{-1}) + B(-z)B(-z^{-1}) = 2 \tag{2}$$

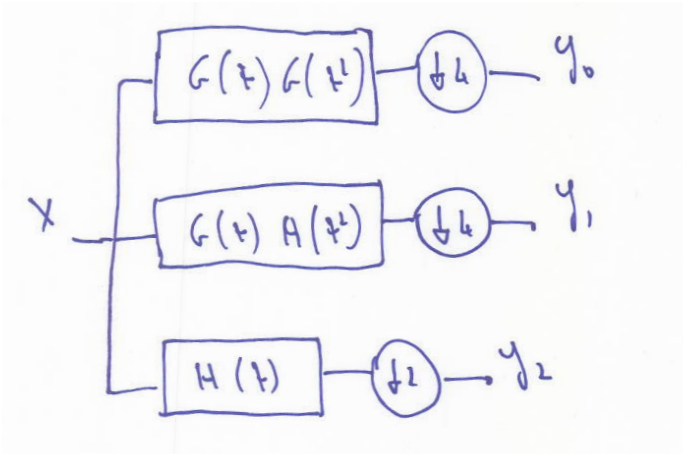
(c) i. The block diagram of the system is:



ii. The equivalent system is



(d) i. The equivalent single-level three-channel filter bank is given below:



The filters are:

$$G(z)G(z^2) = \frac{1}{2}(1 + z + z^2 + z^3)$$

$$G(z)H(z^2) = \frac{1}{2}(1 + z - z^2 - z^3)$$

and

$$H(z) = \frac{1 - z}{\sqrt{2}}$$

ii. $y_0[n]$ is non-zero because the equivalent filter doesn't have any zero at $\omega = 0$. Instead $y_1[n], y_2[n]$ are annihilated because the filter $H(z)$ has one zero at $\omega = 0$.

3. (a)

$$Y_0(z) = \frac{1}{2}[H_0(z^{1/2})X(z^{1/2}) + H_0(-z^{1/2})X(-z^{1/2}]$$

and

$$Y_1(z) = \frac{1}{2}[H_1(z^{1/2})X(z^{1/2}) + H_1(-z^{1/2})X(-z^{1/2}].$$

Therefore

$$\hat{X}(z) = \frac{G_0(z)}{2}[H_0(z)X(z) + H_0(-z)X(-z)] + \frac{G_1(z)}{2}[H_1(z)X(z) + H_1(-z)X(-z)].$$

This implies that PR is achieved when

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2$$

and

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0.$$

(b) We have that $G_0(z) = (z + 2 + z^{-1})/(2\sqrt{2})$.

i. Given that $H_0(z) = cz^2 + bz + a + bz^{-1} + cz^{-2}$, we need three linear constraints for a unique $H_0(z)$. We have that

$$P(z) = \frac{1}{2\sqrt{2}}(cz^3+(b+2c)z^2+(a+c+2b)z+2(a+b)+(a+c+2b)z^{-1}+(b+2c)z^{-2}+cz^{-3}),$$

therefore the half-band conditions lead to only two constraints:

$$(a + b) = \sqrt{2}$$

and

$$(b + 2c) = 0.$$

So many $H_0(z)$ can satisfy the half-band conditions.

ii. By imposing that the filter have a zero at $z = -1$, we obtain a third constraint:

$$a + 2c - 2b = 0$$

leading to:

$$a = \frac{3}{4}\sqrt{2}$$

$$b = \frac{\sqrt{2}}{4}$$

and

$$c = -\frac{\sqrt{2}}{8}.$$

So

$$H_0(z) = \frac{\sqrt{2}}{8}(-z^2 + 2z + 6 + 2z^{-1} - z^{-2}).$$

iii. We can get a biorthogonal filter bank and the two filters are: $G_1(z) = z^{-1}H_0(-z)$ and $H_1(z) = zG_0(-z)$

(c) i. The necessary conditions for convergence are satisfied since $H_0(e^{j\omega}) = 0$ when $\omega = \pi$ and $H_0(e^{j\omega}) = \sqrt{2}$ when $\omega = 0$. However, the sufficient conditions for regularity are not satisfied since in this case $N = 2$, $B = 3\sqrt{2}$ and this means that the condition for regularity $B < 2^{N-1}$ is not satisfied.

ii. Since $H_0(z)$ has two zeros at $\omega = \pi$, then the wavelet has two vanishing moments.

4. (a) We require

$$a_0 = x(0) = 0$$

and

$$a_0 + a_1 = x(1) = 1.$$

Consequently, the Lagrange polynomial interpolator is $p_l(t) = t$.

(b) By the projection theorem, we know that if $p_l(t)$ is the orthogonal projection then $\epsilon_s(t) = x(t) - p_s(t)$ must be orthogonal to $V = \text{span}(\{\varphi_1(t), \varphi_2(t)\})$. In our case $p_l(t) = t$ and it is immediately evident that this is not the case since, for example, $\langle e_s(t), \varphi_1(t)\rangle \neq 0$.
Specifically:

$$\langle e_s(t), \varphi_1(t)\rangle = \int_0^1 (\sin(\frac{1}{2}\pi t) - t)dt = 2/\pi - 0.5 \neq 0.$$

(c) i. We have:

$$\langle \varphi_1, \varphi_2\rangle = \langle \varphi_2, \varphi_1\rangle = \int_0^1 tdt = \frac{1}{2}.$$

Moreover, $\|\varphi_1\|^2 = 1$ and $\|\varphi_2\|^2 = \frac{1}{3}$. Consequently,

$$1 = \langle \varphi_1(t), \tilde{\varphi}_1(t)\rangle = a_{1,1}\langle \varphi_1, \varphi_1\rangle + a_{1,2}\langle \varphi_1, \varphi_2\rangle = a_{1,1} + \frac{a_{1,2}}{2}$$

$$0 = \langle \varphi_2(t), \tilde{\varphi}_1(t)\rangle = a_{1,1}\langle \varphi_2, \varphi_1\rangle + a_{1,2}\langle \varphi_2, \varphi_2\rangle = \frac{a_{1,1}}{2} + \frac{a_{1,2}}{3}$$

and $a_{1,1} = 4$, $a_{1,2} = -6$.
Similarly, we have: $a_{2,1} = -6$, $a_{2,2} = 12$.
The two dual-basis functions can then be written as:

$$\tilde{\varphi}_1(t) = 4 - 6t$$

and

$$\tilde{\varphi}_2(t) = 12t - 6$$

ii. A.

$$\langle x(t), \tilde{\varphi}_1(t)\rangle = 4\int_0^1 \sin(\frac{1}{2}\pi t)dt - 6\int_0^1 t\sin(\frac{1}{2}\pi t)dt = \frac{8(\pi - 3)}{\pi^2}$$

$$\langle x(t), \tilde{\varphi}_2(t)\rangle = 12\int_0^1 t\sin(\frac{1}{2}\pi t)dt - 6\int_0^1 \sin(\frac{1}{2}\pi t)dt = \frac{12(4 - \pi)}{\pi^2}$$

B.

$$x_v(t) = \sum_{i=1}^{3}\langle x(t), \tilde{\varphi}_i(t)\rangle\varphi_i(t) = \frac{12(4 - \pi)}{\pi^2}t + \frac{8(\pi - 3)}{\pi^2}$$

C. We need to verify that $\langle \epsilon_v(t), \varphi_1(t)\rangle = 0$ and that $\langle \epsilon_v(t), \varphi_2(t)\rangle = 0$. We have

$$\langle \epsilon_v(t), \varphi_1(t)\rangle = \int_0^1 \sin(\frac{1}{2}\pi t) - \frac{12(4 - \pi)}{\pi^2}t - \frac{8(\pi - 3)}{\pi^2}dt = \frac{2}{\pi} - \frac{6(4 - \pi)}{\pi^2} - \frac{8(\pi - 3)}{\pi^2} = 0$$

and

$$\langle \epsilon_v(t), \varphi_2(t)\rangle = \int_0^1 t\left(\sin(\frac{1}{2}\pi t) - \frac{12(4 - \pi)}{\pi^2}t - \frac{8(\pi - 3)}{\pi^2}\right)dt = \frac{4}{\pi^2} - \frac{4(4 - \pi)}{\pi^2} - \frac{4(\pi - 3)}{\pi^2} = 0$$